Niyah St. Onge

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SUMMARY

Environmental Engineering Science student at UC Berkeley (GPA 3.86), graduating May 2028, with a water and environmental focus. Experience producing engineering drawings, calculations and permitting documentation issued for construction, researching the requirements a project has to meet, and writing the technical reports that go to clients. Research background in nearshore hydrodynamics and simulation, plus stormwater and water reuse design on an award-winning competition team. Comfortable working independently on assigned scope and across teams on shared deliverables.

EDUCATION

University of California, Berkeley — College of Engineering *Aug 2024 – May 2028*
B.S. Environmental Engineering Science | Minor: Conservation and Resource Studies | GPA: 3.86 | Dean's List Honoree
Relevant Coursework: Water Chemistry, Environmental Engineering, Fluid Mechanics, Thermodynamics, Engineered Systems and Sustainability, Probability & Statistics, Computer Programming for Engineers, Linear Algebra & Differential Equations, Calculus I–III, Physics I–II, General Chemistry

PROFESSIONAL EXPERIENCE

City Structural Inc. *San Francisco, CA*

Design Engineer Intern *May 2026 – Present*
- Produce conceptual through detailed drawing sets in Layout, SketchUp and AutoCAD that are issued for permitting and construction.
- Perform the supporting design calculations and check every sheet against governing code and ordinance requirements before it is issued.
- Research the code, jurisdictional and permitting requirements each project has to meet, and document where existing conditions fall short.
- Write the technical reports presenting analysis and recommendations to clients, and coordinate with clients through the project.
- Work independently on assigned scope and prioritize deliverables across several concurrent projects on fixed deadlines.

Handshake AI *Remote*

Civil Engineering AI Trainer — Project Neutron *Jan 2026 – Present*
- Author civil engineering and architecture problems used to train frontier AI models, and benchmark model output against expected engineering results.
- Document precisely where and why model output diverges from the expected answer, and use generative AI tools daily to speed up research and drafting.

San Francisco County Transportation Authority *San Francisco, CA*

Data Analyst Intern *May 2022 – Aug 2022*
- Collected and tabulated municipal datasets in Excel, analyzed them for trends, and presented findings to supervisors to inform planning decisions.

WATER & RESEARCH EXPERIENCE

Cal Sustainable Solutions (ASCE Competition Team) *Berkeley, CA*

Water Specialist — Water & Environment Sub-team *Aug 2024 – Present*
- Own the water and environment scope of the annual design proposal, including stormwater system design, water reuse and on-site water balance.
- Modeled a mixed-use development in SketchUp, calculated its building energy and water consumption, and ran material take-offs and cost estimates.
- Coordinate with the other sub-teams against a fixed delivery schedule and present and defend recommendations to judging panels; 2nd Place ASCE Nationals 2025 and 1st Place ASCE MidPac 2026.

Gu Research Group (Prof. Grace Gu) *Berkeley, CA*

Research Assistant *Nov 2025 – Present*
- Model nearshore wave behavior and flow around submerged structures in MATLAB and smoothed-particle hydrodynamics, and report findings to the team.
- Ran the group's first sweep of the wave tank damping boundary, measuring energy absorption across five settings to justify the value used in production runs.

3D Printer Filament Reclamation Project (SERC), UC Berkeley *Berkeley, CA*

3D Printer Filament Reclamation Assistant *Jan 2026 – Present*
- Granulate, melt and extrude failed 3D print scraps into new PLA filament as part of a closed-loop recycling system being developed for campus makerspaces.

Design and Innovation for Sustainable Cities (DISC), UC Berkeley *Berkeley, CA*

Summer Design Cohort *Jun 2025 – Aug 2025*
- Produced the site analysis for a transit-oriented redevelopment proposal, researching the water, environmental and energy systems the site would have to work with, and presented it in formal design reviews.

SKILLS

Design & Drafting: AutoCAD, SketchUp, Layout, Rhinoceros 8, Twinmotion; plan production, drawing review against code requirements, permit documentation
Analysis: MATLAB, Python, SPHinXsys, Microsoft Excel; engineering calculations, fluid and hydrodynamic simulation, cost estimating and material take-offs, statistics
Communication: Client-facing technical report writing, code and permitting research, formal presentations to clients and expert panels, multi-project scheduling